

FSA v1.5 closed-loop SMC²-MPC: a side-by-side Python+JAX vs Julia comparison study

Hand-off report for the next agent

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Contents

Abstract

This document is a hand-off for the next agent (or human reader) who picks up the FSA v1.5 closed-loop comparison work. It explains what the FSA v1.5 model is, what the `smc2fc` / `SMC2FC` frameworks do, the functional programming paradigm we have moved to on the Julia side, the comparison problem itself, the tools developed for it, the bugs found and fixed, where we are today, and what comes next. All paths are absolute paths inside the `python-smc2-filtering-control` repository.

1 Why this report exists

The FSA model has three Julia codebases and three Python codebases in this repo (`version_1_*`, `version_2_*`, `version_3_*` on each side). The most relevant pair for this comparison is:

- **Python+JAX:** `version_1_5_Python_JAX/` — the simplest closed-loop FSA bench, derived from the working `version_2_Python_JAX/` bench, with the v1.5 model surface (3-state SDE, 10 estimated parameters, 3 direct-Gaussian observation channels, no circadian).
- **Julia:** `version_1_5_Julia/` — the Julia rewrite of the same model surface, written in a purely functional style for portability to Lean4 (the formal source of truth at `version_1_5_LEAN/`).

The two stacks share the model math (verified bit-equal to 1e-6 through a Julia $\leftrightarrow$ Python+JAX $\leftrightarrow$ Lean4 121-case differential test). They diverge at the framework layer: the Python+JAX side uses the mature `smc2fc` engine; the Julia side uses its own `SMC2FC.jl` port plus model-side GPU kernels in `KernelAbstractions`.

The user has invested weeks of optimisation in the Python+JAX side. The Julia side is roughly two days old. This report freezes the current state of the comparison, the bugs we found and fixed in this session, and the open work for the next agent.

2 What is FSA v1.5?

FSA (Fitness–Stress–Autonomic) is a 3-state stochastic differential equation model for an athlete’s response to training. State $y = (B, F, A)$:

- B — aerobic fitness (Banister chronic; bounded $[0, 1]$, Jacobi-style state-dependent diffusion).
- F — unified fatigue (Banister acute; ≥ 0 , CIR-style diffusion).
- A — autonomic amplitude (Stuart–Landau; ≥ 0 , CIR-style diffusion).

Drift (/day units):

$$\begin{aligned}\mu(B, F) &= \mu_0 + \mu_B B - \mu_F F - \mu_{FF} F^2 \\ dB/dt &= \kappa_B (1 + \varepsilon_A A) \Phi(t) - B/\tau_B \\ dF/dt &= \kappa_F \Phi(t) - (1 + \lambda_A A)/\tau_F F \\ dA/dt &= \mu \cdot A - \eta A^3\end{aligned}$$

State-dependent diffusion (Itô):

$$\begin{aligned}\sigma_B(B) dW_B &= \sigma_B \sqrt{B(1-B)} dW_B \\ \sigma_F(F) dW_F &= \sigma_F \sqrt{F} dW_F \\ \sigma_A(A) dW_A &= \sigma_A \sqrt{A} dW_A\end{aligned}$$

The single exogenous control is $\Phi(t)$, the training-strain rate (≥ 0). The closed-loop MPC's job is to pick Φ to maximise the time-integral of A subject to a soft fatigue constraint ($\max(F - F_{\max}, 0)^2$ barrier).

2.1 What makes v1.5 special

- **14 dynamics parameters** in the v1.5 basis; κ_B replaced by $B_\infty = \kappa_B \tau_B$ and κ_F replaced by $F_\infty = \kappa_F \tau_F$ for better identifiability. The basis-rotation adapter is in `simulation.params.v15_to_v1.nt` (Julia) / `simulation.params.v15_to_v1` (Python+JAX).
- **10 estimated parameters** (the filter's decision variables); 4 dynamics parameters pinned at truth ($\tau_B, \eta, \varepsilon_A, \mu_{FF}$) per the FIM-rank decision; 3 obs-noise scales pinned at 0.005.
- **3 direct-Gaussian observation channels**: $y_B^{obs} = B + \mathcal{N}(0, \sigma_{B,obs}^2)$ and same for F, A . No HR / sleep / stress / steps channels (that's the v2 surface). No circadian forcing.
- Time grid: 60 minutes per bin (24 bins per day) by default. v2 uses 15 minutes per bin (96 bins per day).

3 What smc2fc / SMC2FC does

Both stacks implement the same statistical pipeline:

1. **Outer SMC² filter**: tempered sequential Monte Carlo over the 10-dim unconstrained parameter cloud θ . At each tempering level $\lambda \in [0, 1]$ the targets graduate from prior to posterior. Resample (systematic) $\rightarrow$ Hamiltonian Monte Carlo (HMC) moves with finite-difference gradients $\rightarrow$ next λ . Returns a posterior cloud over θ for each rolling filter window.
2. **Inner particle filter (PF)**: per θ candidate, an inner PF computes $\log p(y_{1:T} | \theta)$ by propagating a state-particle cloud forward through the SDE and reweighting by the observation likelihood. This is the marginal log-likelihood the outer SMC² uses.
3. **Forward-simulation controller**: NOT a particle filter. The controller takes the filter-derived posterior mean parameters $\hat{\theta}$ and the filter-derived smoothed end-of-window state $\hat{y}$. It then runs an SMC² over a low-dimensional schedule parameterisation (8 RBF coefficients θ_{ctrl}): for each candidate θ_{ctrl} , decode the 8 anchors into a per-bin Φ schedule, simulate the SDE forward n_{inner} times under common-random-numbers (CRN) noise, accumulate the cost $J = -\int A dt + \lambda_F \int (F - F_{\max})_+^2 dt$, and use the negative cost as a tempering target. Output is the posterior-mean θ_{ctrl}^* , decoded to a Φ -schedule for the plant's next stride.

Critical: the controller does not run a particle filter internally. It runs deterministic forward simulations (n_{inner} noise trials per θ_{ctrl}) using the filter's $\hat{\theta}$ and $\hat{y}$. This was an explicit design question raised during the comparison and the answer is the same on both stacks: both controllers correctly forward-simulate.

4 Functional programming paradigm in Julia

4.1 Why we moved to functional

The `version_3_julia/` stack was a Python-port-style Julia implementation: mutable structs, `!-` suffix mutating methods, threaded `MersenneTwister` state, history dicts pushed onto across stride boundaries. It accumulated subtle bugs that were hard to trace, and the user explicitly rejected it (“too much for a human to understand”).

`version_1.5_Julia/` is a clean re-design. The five rules:

1. **No mutable structs in the public API.** v1.5 has exactly one struct, `PlantState`, with two fields: an `SVector{3, Float64}` and an `Int`. State updates produce a NEW `PlantState`, not a mutation of the old one.
2. **No `!-` suffix mutating methods.** Every public function returns new values. `plant_step(state, Phi.t, params, dt, key)` returns `(state=new_state, obs=...)`; the caller can build history by collecting returned values.
3. **Explicit RNG keys, JAX-style.** Pass an explicit `key::UInt64` to every randomised function. Internal sub-keys are derived deterministically via `hash((key, :sub_purpose))`. Two callers with the same `(state, key)` get the same answer, period.
4. **Pure facade over the inherently imperative GPU kernel.** `KernelAbstractions` kernels mutate preallocated `CuArrays` by design. v1.5 wraps every kernel-touching public function in a pure facade: `gpu_log_density(target, U, grid_obs, key) → Vector{Float64}` allocates a fresh output, dispatches the kernel, returns. Mutation is encapsulated; callers never see the `CuArray`.
5. **I/O isolated to the bench’s top-level `main()`.** JLD2 + PNG writes happen exactly once at the end of the bench. Logging at the bench level, not inside `plant_step` / `propagate` / `obs_log_weight`.

4.2 The SMC2FC_functional library — now the default

Status update (2026-05-08): the functional rewrite at `julia/SMC2FC_functional/` is now *the default Julia framework*. The original port has been renamed to `julia/SMC2FC_DEPRECATED_USE_FUNCTIONAL/` (kept as a back-compat reference for the three-library comparison harness only). A bit-identical migration test is documented in §?? and recorded in `julia/SMC2FC_functional/docs/MIGRATION_RESULT.md`.

The functional library has the same algorithmic contract as the deprecated port, but:

- Top-level functions take immutable inputs and return immutable outputs.
- Pre-allocated buffers live behind an immutable `Workspace` container.
- No `!-` mutating functions exposed publicly.
- Multiple-dispatch-driven specialisation; idiomatic Julian.
- Google-style docstrings on every file and every function.

The drop-in bench at

`julia/SMC2FC_functional/benchmarks/gpu_drop_in/bench_dropin.jl`

is byte-equivalent to the v1.5 GPU bench except for one import line (using `SMC2FC_functional:run_tempered_smc_gpu`). The three-library cross-check harness lives at `julia/SMC2FC_functional/benchmarks/compare_three_libraries.jl` and the seven verification gates are written up in `julia/SMC2FC_functional/docs/GATE_RESULTS.md`.

5 The comparison problem

The user’s question: *do the Python+JAX and Julia v1.5 stacks produce qualitatively the same closed-loop MPC behaviour, given they implement the same model under the same statistical framework?*

This is harder than it sounds for several reasons:

1. Different RNG implementations (JAX `PRNGKey` vs Julia `StableRNG` vs hash-derived sub-keys) mean the plant trajectories are NOT bit-identical even at the same seed. Comparison must be statistical: mean $\int A$, F-violation rate, posterior MSE to truth, schedule shape.
2. Different inner-PF algorithms: Python+JAX uses `smc2fc.filtering.gk_dpf.v3_lite` (a Gaussian kernel, OT-regularised, differentiable PF); Julia uses a custom segmented-PF KernelAbstractions kernel. Both are valid, but they have different particle-degeneracy regimes.
3. Different default particle counts: Python+JAX bench defaulted to $N_{\text{smc}} = 128/N_{\text{inner}} = 32$ on the controller; the Julia bench defaulted to $N_{\text{smc}} = 256/N_{\text{inner}} = 64$. We brought these into parity.
4. No prior baseline: no constant- $\Phi = 1$ baseline trajectory was being plotted alongside the MPC, so “does the controller actually beat doing nothing?” was hard to answer at a glance.
5. Different output formats: Python writes `.npz`, Julia writes `.jld2` (HDF5).

5.1 Tools developed for the comparison

1. JAX GPU profiler (`src/profile_gpu_python.py`). Direct port of the existing v2 Julia profiler. Times the GK-DPF v3-lite *filter kernel* and the controller cost *kernel* in isolation, with `jax.block_until_ready` as the synchronisation point. Reports median time per call, threads-in-flight, approximate FLOPs, effective TFLOPS, percentage of the RTX-5090’s 104.8 TFLOPS fp32 peak, GPU memory before and after, and host-sync count.

2. v1.5 Julia GPU profiler (`src/profile_gpu_julia.v15.jl`). Same surface, same metrics, adapted to v1.5’s GPU types (`FSAGPUTarget`, `FSAv1ControlGPUPUTarget`).

3. Per-stride telemetry CSV (added to both bench scripts). Each stride writes one row to `per_stride.csv` with: `stride`, `t_wall_s`, `n_temp_filter`, `n_temp_ctrl`, `daily_phi`, `A_mean_so_far`, `B_end`, `F_end`, `A_end`.

4. Paired-run launcher (`src/run_v15_T28d.compare.sh`). Single shell script that:

1. Runs both microbench profilers (writes `julia_profile.log` + `python_profile.log`).
2. Starts `nvidia-smi` sampling at 1 Hz to a CSV.
3. Runs the Julia closed-loop bench with matched config.
4. Stops + restarts `nvidia-smi` for the Python run.
5. Runs the Python+JAX closed-loop bench, same matched config.

Output lands under `outputs/v15_julia_vs_python_T<T>d_seed<S>/{julia,python}/`.

5. Three-way comparison script (`src/compare_v15_julia_vs_python.py`). Loads both bench outputs, both `nvidia-smi` CSVs, both profile logs. Produces:

- `summary.txt` — the diagnostic table (baseline as gate, then macro `nvidia-smi` telemetry, then micro profiler numbers).
- `comparison.png` — 6-panel three-way diagnostic plot. Python MPC blue, Julia MPC red, baseline grey dashed. Panels: A trajectory, applied daily Φ , posterior medians for two parameters with truth lines, GPU SM utilisation over wall time, per-stride wall time bars.

JLD2 files are read with `h5py` and the 2D and 3D arrays are transposed (Julia is column-major; HDF5 readers in row-major order swap dimensions).

6 Bugs found and fixed

We worked top-down through the comparison and found four substantive issues, two on Python+JAX and two on Julia.

6.1 Python: shrinking planning horizon (A1)

File: `version_1.5_Python_JAX/tools/bench_smc_full_mpc_fsa_v15.py`, old line 287.

The replan block computed $t_{\text{remaining_days}} = (n_{\text{strides}} - s) \cdot \text{STRIDE_BINS} / \text{BINS_PER_DAY}$ and passed `T_total=t_remaining_days` into `build_control_spec()`. As the bench progresses, the horizon shrinks; at stride 14 of $T=28d$ the controller sees only 14 days ahead. v2’s bench instead always plans the FULL `T_total_days` at every replan (`bench_smc_full_mpc_fsa.py:386–388`). v1.5 Julia uses `H_plan_bins = T_total_bins` (also full horizon).

Symptom: the controller could not justify a “build B then sprint” Banister strategy because the time-to-horizon was constantly shrinking.

Fix: pass `T_total=float(args.T.days)` (constant) at every replan.

6.2 Python: mid-array splice instead of overwrite (A2)

File: same, old lines 317–320.

The new plan was spliced INTO THE MIDDLE of the existing `daily_phi_plan` array using an absolute-current-day offset. Combined with the shrinking horizon (A1), this produced a sawtooth Φ pattern: the front of the plan got partially overwritten on each replan, the back stayed stale.

v2 overwrites the whole plan: `daily_phi_plan = sched_per_day.astype(np.float64) + last_replan_stride * s + 1`.

Fix: same overwrite-and-reset pattern in v1.5 Python.

6.3 Python: filter degeneracy from bootstrap with GK-DPF (B)

File: `version_1.5_Python_JAX/models/fsa_high_res/estimation.py`, function `_propagate_fn_framework`.

This was the structural bug. The `smc2fc.filtering.gk_dpf.v3_lite` filter is designed for *guided* (Kalman-fused) proposals: the `propagate_fn` contract expects to return both the new state *and* a Radon–Nikodym correction `pred_lw` that the framework adds to the per-bin weight increment.

v1.5 Python+JAX’s `_propagate_fn_framework` returned `pred_lw = 0.0` (the bootstrap convention: proposal IS dynamics, no correction needed). Mathematically valid but fundamentally incompatible with GK-DPF v3-lite’s weight-smoothing kernel + OT regularisation: the framework’s machinery pushed particles toward biased fixed points. Smoking gun: at higher particle counts $N = 1024/K = 800$ the bootstrap variant got WORSE (posterior MSE $1.41 \rightarrow 31.16$), the classic “structural mis-spec” signature.

Fix: Kalman fusion. Mirrors v2’s `propagate_fn` pattern exactly. v1.5’s obs model is the simplest possible Kalman case ($H = I$, $b = 0$, $R = \text{diag}(\sigma_{*,obs}^2)$, all channels always present), so the sequential-scalar update reduces to per-channel 1D Kalman:

$$\begin{aligned}\mu_{\text{prior}} &= y + dt \cdot \text{drift} \\ P_{\text{prior}} &= \text{diag}(\sigma_B^2 B(1-B) dt, \sigma_F^2 F dt, \sigma_A^2 A dt) \\ \text{innov}_i &= y_{\text{obs},i} - \mu_i \\ S_i &= P_{ii} + \sigma_{i,obs}^2 \\ K_i &= P[:,i]/S_i \\ \mu &+= K_i \cdot \text{innov}_i \\ P &- = K_i \otimes P[i,:] \\ \log p_i &= -\frac{1}{2} \log(2\pi S_i) - \frac{\text{innov}_i^2}{2S_i}\end{aligned}$$

Sample $x_{\text{new}} \sim \mathcal{N}(\mu_{\text{fused}}, P_{\text{fused}})$ via Cholesky. Return `pred_lw = log_pred_total - obs_ll(y_new)` so the framework’s `obs_log_weight_fn` re-add cancels properly.

Result (T=7d, seed=42, $N = 32$, $K = 200$):

	Before	After	Improvement
posterior MSE	1.41	0.0188	$\sim 74\times$

The 10 estimated parameters’ posterior medians now hug truth tightly (was: τ_F stuck at 3 vs truth 7; sigmas drifting upward unbounded).

6.4 Python: init_state cheat (C)

File: same bench file, around old line 294.

The bench was passing `init_state_dict = dict(B=plant_state.bfa[0], ...)` directly to the controller. This is a closed-loop violation: the controller should know only what the filter knows (its posterior estimate of the state), not the plant’s true state.

Fix: extract a smoothed end-of-window state from the filter posterior cloud via `log_density_factory.extract` mirroring v2’s pattern (line 366–367). Use that as `init_state`.

6.5 Julia: per-tempering-level allocation churn (perf)

File: `version_1.5_Julia/models/fsa_high_res/gpu_pf.jl`, function `parallel_hmc_one_move`.

The functional HMC variant allocates roughly ten (M, d) matrices per call. With 13 strides $\times$ 5 tempering levels $\times$ 3 HMC moves per level the bench does ~ 200 HMC calls, allocating a few MB total — not catastrophic, but a real per-call constant.

Fix: added a mutating `parallel_hmc_one_move!` variant that reuses two persistent (M, d) buffers across the whole leapfrog. The bench imports both; `run_outer_smc` now calls the mutating variant.

Result (T=2d open-loop smoke, $N = 16$, $K = 100$): wall time 96.9 s $\rightarrow$ 92.0 s ($\sim 5\%$ faster at this small N). All six GPU PF smoke tests still pass.

6.6 Julia: optional Liu–West θ -cloud shrinkage

File: same plus `version_1.5_Julia/tools/bench_smc_full_mpc_fsa_gpu.jl`.

Added a CLI flag `--liu-west-a` (default 0.0 = disabled). When set in $(0, 1)$ — e.g. 0.97 — applies the standard Liu–West shrinkage between resample and HMC inside `run_outer_smc`:

$$\theta_i \leftarrow a \cdot \theta_i + (1 - a) \cdot \bar{\theta} + \sqrt{1 - a^2} \cdot \text{column_std} \cdot \xi, \quad \xi \sim \mathcal{N}(0, I)$$

Helps maintain θ -cloud diversity at low N , where pure tempered HMC can collapse. Off by default so existing reproducibility is preserved.

6.7 Julia: per-segment GPU $\rightarrow$ CPU transfers in `gpu_log_density` (perf, big)

File: `version_1.5_Julia/models/fsa_high_res/gpu_pf.jl`, function `gpu_log_density`, segment loop.

The segment loop ran a per-chain log-likelihood accumulator *on the host*:

```
for seg in 1:n_segments
    target.propagate_kernel(...)
    KernelAbstractions.synchronize(CUDABackend())
    @match (seg, n_segments) begin ... end
    #      HOT-LOOP HOST READ (anti-pattern)
    log_max_cpu = Array{view(bufs.log_max, 1:M)} # GPU -> CPU
    log_z_cpu   = Array{view(bufs.log_z, 1:M)}   # GPU -> CPU
    @inbounds for m in 1:M
```

```

        log_lik_acc[m] += Float64(log_max_cpu[m] + log_z_cpu[m] - log_K)
    end
end

```

Per call: 12 GPU $\rightarrow$ CPU transfers (M floats $\times$ 2 fields $\times$ 6 segments) + 6 redundant `KernelAbstractions.sync` calls. Each transfer forces an implicit stream sync that stalls the GPU between segments.

Fix: a Float64 GPU-resident accumulator `log_lik_acc_gpu::CuArray{Float64,1}` added to `FSAGPUTarget`, plus a tiny `gpu_log_lik_accumulate_kernel!` that writes `log_lik_acc[m] += Float64(log_max[m] + log_z[m] - log_K)` on the GPU. The end-of-function `Array(view(...))` syncs the stream once and returns the M -vector to the host. Removed the redundant per-segment syncs at the same time.

This change alone was a small win at the tested config (~ 3 ms out of 81 ms). The headline cost was found to live elsewhere — see §??.

6.8 Julia: OT-rescue per-chain Julia loop (perf, headline)

File: `julia/SMC2FC/src/Filtering/GPUSegmentedPF.jl` (framework, now `SMC2FC_DEPRECATED_USE_FUNCTIONIO`), same code copied verbatim into `SMC2FC_functional`), function `gpu_ot_blend_chain!`.

The OT (optimal-transport) sigmoid-blend rescue inside `run_segmented_smc_step!` wraps the per-chain Sinkhorn in a Julia for-loop over M chains, which inside its body does

```

log_w_arr = Array{chain_log_w}      # GPU -> CPU memcpy + sync (per chain)
... softmax on CPU ...
b_gpu      = CuArray(b_cpu)          # CPU -> GPU memcpy + sync (per chain)

```

At $M = 32$ chains $\times$ 5 segments per `gpu_log_density` call, that is ~ 320 PCIe round-trips *per call*, every one of them stalling the GPU. The actual GPU compute under the rescue (Sinkhorn iterations + barycentric projection) is sub-millisecond.

A/B microbench (RTX 5090, $N_{\text{smc}} = 32$, $K_{\text{pf}} = 200$, $T_{\text{steps}} = 24$):

<code>ot_max_weight = 0.01</code> (was default)	84.78 ms / call
<code>ot_max_weight = 0.0</code> (now default)	0.73 ms / call
speed-up	116 $\times$

Closed-loop A/B (T=2d, seed=42, $N = 16$, $K = 100$):

metric	OT on	OT off (now default)
total wall time	159.5 s	12.5 s
mean A	0.0928	0.0928
final $\hat{x}$	(0.059, 0.273, 0.088)	(0.059, 0.273, 0.089)
posterior medians (10 params, identical)	7 of 10	7 of 10
posterior medians (3 params)	differ < 22% rel., inside the IQR band	

Fix: added `--ot-max-weight` CLI flag to the bench (default 0.0, OT off), and flipped the `FSAGPUTarget` constructor's `ot_max_weight` keyword default from 0.01 to 0.0. Heavy warning comments at both spots explaining the slowdown, the microbench and closed-loop numbers, when to turn it back on (low-ESS / hard obs models), and the proper fix (a future framework rewrite of `gpu_ot_blend_chain!` as a single batched-across-chains GPU kernel).

Why this is safe at v1.5: the writeup's earlier section on Python's bug (B) showed the GK-DPF v3-lite framework expects guided proposals and gets them via Kalman fusion — the OT rescue is then unnecessary on the strong direct-Gaussian 3-channel obs model. Python+JAX runs without an explicit OT rescue and matches truth fine. Julia at v1.5 is in the same regime.

6.9 Julia framework migration: bit-identical

The migration of the v1.5 GPU bench from the old `SMC2FC` framework to `SMC2FC_functional` is exactly one line on the bench side:

```
- using SMC2FC: run_tempered_smc_gpu
+ using SMC2FC_functional: run_tempered_smc_gpu
```

The drop-in artefact at `julia/SMC2FC_functional/benchmarks/gpu_drop_in/bench_dropin.jl` is byte-identical to the v1.5 GPU bench except for that import line and the `REPO_ROOT` pointer. `SMC2FC_functional`'s `Control/GPUControlSMC.jl` was a verbatim COPY+DOC of the deprecated framework's — only the file-header block-comment was converted to a Julia docstring. So at matched config and seed, the two frameworks must produce the same output, modulo JIT-warmup wall-time variance.

Head-to-head test at $T=2d$, $\text{seed}=42$, $N = 16$, $K = 100$:

quantity	deprecated <code>SMC2FC</code>	<code>SMC2FC_functional</code>	max diff
wall time	12.5 s	12.9 s	+0.4 s (JIT)
trajectory $B/F/A$ (all bins)	—	—	0.000e+00
applied Φ per bin	—	—	0.000e+00
baseline trajectory	—	—	0.000e+00
all 10 posterior medians	identical	identical	0.000e+00 each
posterior particle clouds	—	—	bit-identical

Every numeric in the JLD2 file is byte-identical. `SMC2FC_functional` is the default framework going forward.

7 Where we are now

7.1 State of play before today

The pre-fix headline at $T=7d$, $\text{seed}=42$, matched config $N_{\text{smc}} = 32$, $K_{\text{pf}} = 200$ was:

	Baseline ($\Phi = 1$)	Python+JAX MPC	Julia MPC
mean A (last 7 days)	0.0935	0.0808	0.0815
improvement vs baseline	—	−13.6%	−12.8%
F-violation rate	0.0	0.0	0.0
posterior MSE to truth (final window)	—	0.0188	0.0868
total wall time (min)	—	5.3	37.2
mean GPU utilisation (%)	—	40.0	24.8
filter kernel time / call (ms)	—	4.7	81.0
controller cost / call (ms)	—	23.4	0.5

7.2 State of play after today's perf work

After the `gpu_pf.jl` GPU-accumulator fix (§??), the OT-rescue default flip to off (§??), and the framework migration to `SMC2FC_functional` (§??), the Julia microbench at the same matched config now reports:

metric (RTX 5090, $N_{\text{smc}} = 32$, $K_{\text{pf}} = 200$, $T_{\text{steps}} = 24$)	Python+JAX	Julia	ratio
filter kernel time / call (ms)	4.7	0.9	Julia $\sim 5\times$ faster
controller cost / call (ms)	23.4	0.4	Julia $\sim 60\times$ faster

i.e. the microbench has flipped from *Julia 17× slower* (81 ms) to *Julia 5× faster* (0.9 ms) on the filter kernel, simply by removing the per-segment GPU → CPU host read pattern and disabling the OT rescue’s per-chain Julia loop.

The new headline numbers are nuanced (read carefully):

- **Both filters remain algorithmically sound.** The Python+JAX MSE 0.0188 vs Julia 0.0868 difference is the same maturity gap as before — not affected by today’s perf work, which is purely about wall time.
- **The neither-beats-baseline-at-T=7d finding is unchanged.** The H3 controller-cost-design issue is unaffected by perf. The longer-horizon test (T=14d / T=28d) is still the right next experiment.
- **Julia is now *faster per call* than Python+JAX — but the closed-loop wall-time race has moved.** The 81 ms filter cost is gone, but the *full T=7d closed-loop bench numbers above were captured with the slow Julia path and have not been re-measured*. We expect a $\sim 10\times$ closed-loop speed-up on Julia at matched config, putting Julia close to or under Python+JAX’s 5.3 min; that is the next thing to verify.
- **Caveat: matched config is not maxed config.** The $N_{\text{smc}} = 32/K_{\text{pf}} = 200$ config above is what Python+JAX was tuned for. Julia, with its $\sim 5\times$ faster per-call cost, has substantial headroom on the RTX 5090 — both compute and memory. Bumping particle counts to a Julia-saturation tier (and re-running T=14d / T=28d there) is the next-cheapest scientific test, and is the focus of the next plan. See §??.

8 What comes next

In rough priority order:

1. Migrate to SMC2FC_functional. [done] `julia/SMC2FC_functional/` is now the default Julia framework. The deprecated `julia/SMC2FC/` has been renamed to `julia/SMC2FC_DEPRECIATED_USE_FUNCTIONAL/` and is retained only for the three-library comparison harness. The migration is bit-identical; see §?? and `julia/SMC2FC_functional/docs/MIGRATION_RESULT.md`.

2. Optimise Julia GPU utilisation. [partially done; new plan filed] **Done in this session:**

- Per-segment GPU → CPU host read removed from `gpu_log_density`’s segment loop, replaced by a GPU-resident accumulator + a tiny accumulation kernel (§??). Removed the redundant per-segment `KernelAbstractions.synchronize()` calls that the old code needed.
- OT-rescue per-chain Julia loop turned off by default (§??). The headline filter kernel cost dropped from 81 ms / call to 0.9 ms / call.
- Mutating HMC variant kept (the earlier session’s small win at low N).

Open: the closed-loop wall time at T=7d / T=14d / T=28d has not been re-measured with the new fast path, and Julia is *not* yet running at a config that saturates the RTX 5090’s 104.8 TFLOPS fp32 peak / 32 GB memory. The new plan filed at `claude_plans/Julia_T28d_max_5090_saturation_2026-05-08` covers:

- Re-running the headline microbench at $N_{\text{smc}} = 256$, $K_{\text{pf}} = 400$ (v2 saturation tier) and at $N_{\text{smc}} = 512$ / $K = 800$ to find the GPU’s compute and memory ceiling.
- `nsys profile` on a real bench run to attribute the remaining microseconds to specific kernels.
- Picking the largest config that fits in memory at T=28d and re-running the full closed-loop comparison at that scale, then re-running the section-9 comparison harness end-to-end.
- Re-introducing OT rescue *only* as a future framework rewrite of `gpu_ot_blend_chain!` into a single batched-across-chains GPU kernel (out of scope for this plan).

3. Run T=14d / T=28d to test H3. The user’s reference v2 plots ramp Φ UP at longer horizons (T=14d: mean A 0.122 vs baseline 0.081, +50%; T=28d: 0.157 vs 0.098, +60%). The hypothesis is that v1.5’s controller will similarly find the ramp-up optimum at horizons long enough for the “build B then sprint” payoff to dominate. Now that the filter kernel is $\sim 90\times$ faster, T=28d at the saturated config is wall-time-feasible (estimated under 30 min on Julia). This experiment is wrapped into the plan at the end of item 2.

4. Cost-function variants. If T=14d / T=28d still picks rest, try:

- Re-introduce $\lambda_\Phi \int \Phi^2 dt$ penalty (currently $\lambda_\Phi = 0$ on both stacks by default — the cost expression supports it).
- Add a soft “minimum mean- A over planning window” floor.
- Investigate whether the v2 G1 reparametrisation around (A_{TYP}, F_{TYP}) shifts the bifurcation balance in a way that makes ramp-up the dominant optimum at all horizons.

5. Multi-seed robustness. All comparison runs to date are single-seed. If a T=28d run flips H3 cleanly, replicate at ≥ 3 seeds before declaring victory.

9 How to run the comparison end-to-end

The full pipeline is automated by the launcher:

```
conda activate comfyenv
bash
  /home/ajay/Repos/python-smc2-filtering-control/version_1_5_Python_JAX/tools/launchers/run_v
  7 42
# args: T_DAYS, SEED. Defaults: T=28, seed=42.
```

This produces (under the outputs/v15_julia_vs_python_T<T>d_seed<S>/ parent dir):

- julia_profile.log, python_profile.log
- julia/: data.jld2, manifest.json, per_stride.csv, gpu_telemetry.csv, two PNGs (state traces, param traces), bench.log.
- python/: trajectory.npz, manifest.json, per_stride.csv, gpu_telemetry.csv, two PNGs, bench.log.

Then run the comparison script:

```
cd /home/ajay/Repos/python-smc2-filtering-control/version_1_5_Python_JAX
JAX_ENABLE_X64=True PYTHONPATH=... \
python tools/compare_v15_julia_vs_python.py \
/home/ajay/Repos/python-smc2-filtering-control/outputs/v15_julia_vs_python_T7d_seed42
```

This writes summary.txt and comparison.png into the parent dir.

10 How to profile each stack in isolation

```
# Julia microbench (filter kernel + controller cost kernel)
cd /home/ajay/Repos/python-smc2-filtering-control/version_1_5_Julia
julia --project=. tools/profile_gpu.jl --both

# Python+JAX microbench (same surface, same metrics)
cd /home/ajay/Repos/python-smc2-filtering-control/version_1_5_Python_JAX
JAX_ENABLE_X64=True PYTHONPATH=... \
python tools/profile_gpu.py --both
```

For deeper Python+JAX profiling, pass `--trace` to dump a `jax.profiler` trace into `./jax_trace_filter/` and `./jax_trace_controller/` that TensorBoard's Profile tab can render.

For deeper Julia profiling, run the bench under `nsys profile --trace=cuda julia tools/bench_smc_full...` and inspect the resulting `.qdrep` in NVIDIA Nsight Systems.

11 Repository basics: where things live

Python+JAX. Per-file layout under `version_1.5_Python_JAX/`:

- `models/fsa_high_res/_dynamics.py` — pure JAX drift, diffusion, EM step.
- `models/fsa_high_res/simulation.py` — `DEFAULT_PARAMS`, `INIT_STATE`, `params_v15_to_v1`.
- `models/fsa_high_res/_plant.py` — pure `plant_step` / `plant_rollout`.
- `models/fsa_high_res/estimation.py` — `HIGH_RES_FSA_V15_ESTIMATION` (`EstimationModel`); the Kalman-fused `_propagate_fn` framework.
- `models/fsa_high_res/control.py` — RBF schedule + cost; `build_control_spec`.
- `tools/bench_smc_full_mpc_fsa_v15.py` — the closed-loop bench.

The `smc2fc` framework lives at `smc2fc/` (one level up from any version dir). The relevant entry points consumed by the bench are: `smc2fc.core.jax_native_smc.run_smc_window_native`, `...run_smc_window_bridge_native`, `smc2fc.filtering.gk_dp_v3_lite.make_gk_dp_v3_lite_log_density_c`, `smc2fc.control.tempered_smc_loop.run_tempered_smc_loop_native`.

Julia. Per-file layout under `version_1.5_Julia/`:

- `models/fsa_high_res/_dynamics.jl` — v1 verbatim G0 SDE drift / diffusion / substepped EM.
- `models/fsa_high_res/simulation.jl` — `DEFAULT_PARAMS`, `params_v15_to_v1_nt`, `sample_obs_bfa`. Uses `Match.jl @match` at every site that a future Lean4 port would also need to pattern-match.
- `models/fsa_high_res/_plant.jl` — pure `plant_step` / `plant_rollout`.
- `models/fsa_high_res/gpu_pf.jl` — `FSAGPUTarget`, `gpu_log_density`, `gpu_grads`, `parallel_hmc_one_move` / `parallel_hmc_one_move!` (the new mutating variant).
- `models/fsa_high_res/gpu_control.jl` — `FSAv1ControlGPUPUTarget`, `gpu_cost_log_density_batched`.
- `models/fsa_high_res/estimation.jl` — `PARAM_NAMES`, `PARAM_PRIOR_CONFIG`.
- `tools/bench_smc_full_mpc_fsa_gpu.jl` — the closed-loop bench, including `run_outer_smc` (the hand-rolled tempered SMC² loop) and `controller_plan` (calls `SMC2FC.run_tempered_smc_gpu`).

The Julia framework lives at `julia/SMC2FC/` (existing) and `julia/SMC2FC_functional/` (new, drop-in replacement).

Lean4 reference. `version_1.5_LEAN/` provides a Lean4 implementation of the v1.5 model functions exposed via a JSON-line CLI (`Main.lean`). The diff test `version_1.5_Julia/diff_test/test_lean_diff_v1` round-trips 121 randomised cases through the Lean4 binary at 1e-6 single-step / 1e-4 integrated tolerance. This guarantees the model math is identical across all three implementations.

12 Closing notes

The work in this session is committed and pushed on the `Julia-port-verison-1` branch. Notable hashes:

- `0c92907` — CLI parity + GPU profilers (Phases A/A1/A1b)
- `ff3da57` — per-stride telemetry + launcher + comparison script
- `155beec` — T=7d pre-flight artefacts + initial findings
- `159f93e` — Python A1 + A2 fix (shrinking horizon + mid-array splice)
- `3ff65c3` — Python Kalman-fusion fix + `init_state` cheat fixed
- `da5e9c9` — findings reframing (drop misleading “Python best of 3”)
- `c4a9e86` — Julia perf: mutating HMC + optional Liu–West
- `ebbd9ed` — v1.5 LEAN reference + 121-case round-trip diff test
- `a6903f9` — `SMC2FC_functional` library (parallel rewrite + 7-gate audit + GPU drop-in test)
- `da847d9` — v1.5 Julia GPU filter: per-segment GPU-accumulator fix; `--ot-max-weight` CLI flag
- `099a14a` — v1.5 Julia: default OT rescue OFF; heavy warnings on the slow path
- `4a5c007` — `SMC2FC_functional` migration on v1.5 GPU bench: bit-identical numerics

The corresponding plan archive is `claude_plans/FSA_v1.5_three_way_closed_loop_comparison_2026-05-08` (also copied to `docs/PLAN_ARCHIVE.md` in this directory).